Cosmic Queries – Light Sails & Quantum Scales

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis

This episode of Star Talk features a Q&A session where astrophysicist Neil deGrasse Tyson and co-host Chuck address fan-submitted inquiries ("Cosmic Queries") on the intersection of observational cosmology, theoretical physics, and scientific jargon. The core thesis explores the boundaries of human knowledge regarding the multiverse, time dilation, space exploration capabilities, and historical astronomical observations. Tyson emphasizes a shift from classical intuition to relativistic reality, using analogies like rubber sheets and kilns to clarify complex concepts such as general relativity, while simultaneously critiquing statistical illiteracy in both science and society.

2. Key Arguments & Data

- **Multiverse & The "Other":**
  - One version of the multiverse consists of bubbles expanding within a grid of spacetime; these bubbles never overlap unless there is a measurable impingement.
  - Detection relies on analyzing Cosmic Microwave Background (CMB) fluctuations for statistical differences that might indicate an impinging universe.
  - To date, no impinging signatures have been found in the statistics of the CMB.
  - Laws of physics (e.g., electron charge) likely vary between universes, making inter-universal travel dangerous due to atomic collapse or disintegration.
- **Astrophysical Jargon:**
  - Neil critiques words that obscure meaning with too many syllables, such as "magnetohydrodynamics."
  - Historical terminology can be misleading: The term "Planetary Nebula" is incorrect because they are gaseous structures, not planets, based on their appearance rather than nature.

    - Preference: Neil lists words that mean exactly what they say (e.g., sunspots, thermonuclear fusion) and dislikes jargon that acts as a barrier to understanding.
- **Space Exploration & Propulsion:**
    - Project Starshot utilizes postage-stamp-sized spacecraft accelerated by gigawatt Earth-based lasers.
    - Calculations suggest these craft could reach speeds of 20% the speed of light (c) to Alpha Centauri (4 light-years away), arriving in ~20 years.
    - Traditional rockets with high acceleration would require 50,000+ years to reach Alpha Centauri. The New Horizons mission to Pluto set the record for the fastest launched spacecraft but is still significantly slower than Project Starshot's theoretical speeds.
- **Relativity & Time:**
    - There is no absolute "now" or simultaneity in the universe due to relativistic principles.
    - Time dilation affects distant observations; supernovae from when the universe was half its current size have stretched light curves, providing direct evidence that time passes differently than local measurements.
    - The speed of light remains constant regardless of the observer's position or time.
- **Gravity Visualization:**
    - The "rubber sheet" analogy (mass on a trampoline) is flawed because it implies mass acts upon an external space-time fabric, whereas in reality, all objects possess their own deformed space-time structure.
- **Statistical Significance:**
    - Statistical analysis revealed that stars do not appear randomly; there is a significant clustering of close pairs and triplets compared to random scattering.
    - This statistical approach refutes "hot hand" fallacies, showing that clumping is expected by chance but requires data validation (like gravity lensing in Hoag's Object) to confirm coherence over coincidence.
- **Fractals in Nature:**

    - Fractals are aesthetically pleasing and found in some botanical structures (e.g., tree branching), but they do not scale well for scientific modeling across different physical regimes (e.g., weather vs. quantum physics).
    - Modeling trees with fractals often fails because the rules at smaller scales differ from larger ones; currently, fractals are viewed as entertainment rather than utility in decoding nature.
- **Early Universe & CMB:**
    - The Cosmic Microwave Background originated when the universe was ~3,000 degrees Kelvin, emitting visible light that covered the entire sky (a "heavenly glow").
    - This radiation redshifted over time due to cosmic expansion from microwaves; at formation, there was no distinct star or planet formation.
    - Analogy: Immersing a room in kiln heat (approx. 3000 degrees) causes all objects in the room to glow with the same temperature, rendering them invisible against the background.

3. Actionable Takeaways

- **Investigate Project Starshot:** Visit official Google pages or X Prize databases for detailed specifications on future solar sail propulsion technology and its feasibility compared to traditional rocketry.
- **Study Statistical Illiteracy:** Apply statistical probability models (like those used to analyze star clustering in Hoag's Object) to everyday observations to avoid assuming magical patterns where only random chance exists.
- **Update Astrophysical Jargon Knowledge:** Read Neil deGrasse Tyson’s book *Universe Down to Earth* or *Lost in Space* to learn about the history of nomenclature (e.g., why "Planetary Nebula" was named) and terms that sound like romance novels (Lagrangian points) rather than technical concepts.
- **Visualize Gravity Correctly:** Discard the rubber sheet analogy as a standalone tool; instead, acknowledge that all matter distorts space-time equally regardless of size or density.
- **Analyze Light Curves:** Recognize that time is not constant; observe distant supernovae to witness light curves stretched by cosmic

expansion, proving relativistic effects on past events.

- **Appreciate Cosmic History:** Understand the visible light era of the universe (3000 K) through analogies of infrared kilns or heat-illusion films (e.g., *The Thomas Crown Affaire*) where temperature differences cause objects to become invisible within a background environment.

Full Transcript

Chuck, that was a fun Cosmic Queries, grab back. My favorite was the person who asked, what would the Cosmic Microwave background be like when it happened? It's like, whoa. Take cover, is this lesson there? You know what Neil told him? Hey man, I was there when it happened and I'll give you a first-hand account. Coming up on Star Talk. Welcome to Star Talk. Your place in the universe where science and pop culture collide. Star Talk begins right now. This is Star Talk, Cosmic Queries. Poporee, crab bag edition. Chuck, how you doing, man? Doing great, Neil. That's right. All right. I like the way you say Poporee. Let me hear it. The Poporee, which is my version of galactic gumbo.

Yeah, we don't we don't know what to call this other than the grab bag. Grab bag seems to work. Yeah, that works. And it's a fan favorite because people ask whatever the hell they want. What you want? All right, Chuck. So what's your first question? All right. Here we go. Tricia Lynch says, hello, Dr. Tyson and Lord Nice. This is Tricia from Beaverton, Oregon. If the multiverse exists. How would we be able to tell where our universe ends and another universe? Begins. Well, thank you. I like that. Well, there's several variations on the you on the multiverse and I only know a couple of them But they're more than the number that I know and they come out of sort of the mathematics of the early universe as you shotgun marry quantum physics and general relativity with string theory and other sort of creative ideas in the early universe. So one version of the multiverse is there's one enormous grid of space time and we're one bubble within that grid. Okay. Okay. The difference is these bubbles are separated from each other.

in a way that you cannot access the other bubble. The end of our universe is wherever that is, but it would not overlap with another universe, even though all of these universes are expanding. If it's a bubble in the spherical sense of a bubble, then the universe ends where the universe begins. Like where is the end of a bubble? Where is the end of a bubble? So you might say that the edge of the bubble is where you begin your journey into the bubble universe. Exactly. Is that an end or a beginning? Is that an end or a

beginning, right? I like that. I like that. Yeah. A legit reinterpretation of the information. So all these universe are out there.

expanding and not overlapping. By the way, on the possibility that they might have overlapped, there are experiments you can do with the cosmic microwave background. What you can do is you can look to see if one part of the cosmic microwave background looks statistically different from another part of the cosmic microwave background in a different direction. And that could be somebody else's universe impinging on ours. And that would show up in the statistical analysis of the microwave background. And people have done that experiment and looked for these signatures and haven't found them. They found, yeah, there's no, there's nobody impinging on our statistical on the statistics of our universe. And I say statistics of because the microwave background is, is a is a tapestry of fluctuations of temperature that existed in the early universe that is imprinted in microwaves that we can observe today. And if there's a different universe, it might have a different imprinting. And you'll be able to see that. You should be able to see that if there was some overlap. And we don't. So that's one kind of multiverse. Another one is just a whole bubble separate from this grid of spacetime.

that I presumably embedded in a higher dimension and it's expanding and we're expanding and never the twain will meet. So, plus it's possible, in fact, it's likely given quantum physics that the laws of physics are slightly different in one universe relative to another. And that would be extremely dangerous if you wanted to visit. the other universe, because you don't know if it's the charge on the electron different. All your atoms fly apart or they collapse. Hey, look who showed up. It's the goo people. Where did these goo people come from? Did you collapse into a pile of goo because all the forces that otherwise kept you alive are doing something different. So, yeah. So that's the best I can address this.

What I might do is next time we get Brian Green on we can bring up that topic with him See if he has any more to add to it because he knows he's more nuanced about Elements of the multiverse that could influence or impact the the full answer to that question cool All right, I want to put a star next to that put a star next to that and we'll hold it aside when Brian Green

shows up again flag it for Brian Green Yeah, it's cool. All right, then let's move on to this is Mary Mitchell. And Mary says, Hi, Dr. Tyson, Lord Nice, Mary from Oregon. Two organs in a row. Two organs in a row. I was just in Oregon just a few weeks ago. Okay. In Asheville, Oregon. Asheville. Oh, my God, that is a party town. Well, is it Asheville or Ashland? I don't know.

I don't know. Ashlyn Art, that's the side of the Oregon Shakespeare Festival. Oh, really? And took in some Shakespeare plays, yeah. Oh, I said it was a party town. It's quite the culture town. Exactly. And the Shakespeare Festival was in North Carolina. Ashlyn is in Oregon. Ashlyn is in Oregon. OK. All right. Mary says this, the cosmic query is, for both of you, OK, what is one of your favorite words in astrophysics? Is there a particular word that reflects the structure function of its referent or that it's just fun to say? Alternatively, is there a word in astrophysics that chafes your geeky underbelly? So what's your issue with it? Thanks. OK. I love that. I love that question. Let me tell you why. My second book ever was titled, Universe Down to Earth. Second book ever came out like in the early 90s. And in fact, I've been asked to update it and have it get re-released, which might happen in the next 18 months or so. And I mentioned that because one of the chapters is titled, The Confused Person's Guide.

to astronomical jargon. Wow. And so the entire chapter is just words that might confuse you, and then I rectify that. And the words are grouped in categories, like words that mean exactly what they say, words that are completely confusing, words that sound like romantic destinations, words that are socially and culturally grouped. Wait, you put this out as a book? Well, that was one chapter of a book. Yeah. Oh, okay. Dude, you should make a pocket book, like a little pocket book. Oh, okay. All right. By the way, I want to cut. I'm just saying. Y'all heard it here. So in terms of like romantic words, so I put it in the romantic word section. Your Lagrangian point We have a whole explainer under your Lagrangian points, yes, but if you didn't otherwise know that's that's gonna show up in some romance book I would like very much To make love to your Lagrangian point If only I could hold your Lagrangian point Not only would you freeze. So what's also there is the four telescopes that observe the sun. Okay. There's a special configuration of the optics where the focus of the telescope because the Sun has unlimited brightness, right, compared to what you're normally looking

at as an astronomer. Not unless you look at it directly from the White House. A lot of people don't know, but nobody knows more about looking at the Sun than I do. I could look at it. I stared it down and I won. Stared it to heart at it, the Sun set. Exactly.

Oh, that's a good one. I stared at it so hard. So what happens is you have the main mirror that reflects the light back to a secondary mirror, and then you take that light and beam it into a separate room. And all the optics are configured so that no matter where the sun is in the sky, the focus will land in this singular room. And that room has all the special optics to do this. And it's called the coup d'et focus. The coup d'etre was the description of that kind of optics. But the room where it comes in is called the coup d'etre room. So I thought that had a little romantic. It does have a romantic. Shall we retire to the coup d'etre room? Oh dear. So I also list words that I hate that were not invented by astronomers because we're much more creative and descriptive.

than our physics counterparts. And so there's a branch of study led by physicists in astrophysics. And it's called magnetohydrodynamics. That's just a word with too many syllables. I'm sorry. Magnetohydrodynamics. And so this is the study of magnetic fields and their influence on gases in the universe, whether those gases be gas clouds. or the gas constituents of a star itself. Magneto-hydrodynamics. So then we have words that mean exactly what they say. Exactly what they say. The sun has spots. Guess what we call them. Sunspots. Thank you. The Jupiter has a big red spot. Guess what we call it. Yeah. The Jupiter bedwetting. Yeah, the big red spot. Jupiter's red spot. And so we tend to be very practical minded about that. And there are a lot of nebulae out there, just gas clouds formed by all multiple different ways. Some are gases that have been released by a star in its death. Could be supernova, could be other mechanisms that will release gas. Others are just gas clouds themselves that have been influenced by stars being born within them that evacuates pockets of Because the pressure from newly formed sunlight will do that. Anyhow, when we look at these things, we say, hey, what does that look like? Hey, that looks like a tarantula. We have a tarantula nebula. Hey, that looks like a ring. That's a ring nebula. These are official names of things. You could look these up.

The ring nebula, the tarantula nebula, the lagoon nebula, the owl nebula. Something that looks like two eyes and...

the owl nebula, the eagle nebula that looks like a spread eagle, like on money, you know, on the seal of the president, with the head of the eagle is pointing sideways so you can see the beak, the eagle nebula. So the whole set of nebulae, we just call them like we see them. And that's very different from other fields where they will try to be very precise in their nomenclature, but then it creates a smoke screen for anyone trying to get close to it. like geologists. Like, you know, I was with the geology friend in the woods and I said, that's a pretty rock there. That's ortho-clays felled spar. Okay, I don't like it. Yeah, exactly. That's yeah. Yeah, take it to Star Trek, buddy. Yeah, exactly. So is there anything that chafes my my speedos? I would say magnetohydrodynamics. Because as an educator, I don't want a smoke screen with words. I don't want the word to be the barrier.

I want the idea to be what might be the challenging point about what I'm describing, not the word itself. Right. Right. And so I don't know if I hit all the elements of that. You want to know what's my favorite word? No, you got it all. What shapes you? What did you like? Yeah, you hit them both. Yeah. Yeah. Here's a word that means exactly what it says. Thermo, nuclear fusion. Absolutely. That's another thermo is heat. Nuclear is the nucleus of the atom. Fusion, you're bringing atoms together. Yes. Space fact. Dark matter is 85% of all the gravity we measure in the universe. And we don't know what's causing it. It doesn't interact with our light. It doesn't even interact with itself. Which means it cannot coalesce to form solid objects. Like...

stars or planets or people. If you like that fact, you can find 4,999 more in Lost in Space. 5,000 facts to help navigate the universe. Lost in Space is now available wherever you get your books. By the way, we have words that are left over from previous times, but we never tidied up the nomenclature. The among the nebula there's a kind of nebula called planetary nebula planetary nebula. Oh, oh, it's sad That's why it's not a thing Okay, are you ready? Go ahead. You ready for this? Okay. Okay, if you Have a telescope and you look up at the night sky and you look at a star the star is so far away It's just a point of light even in a telescope even in a

telescope even if they're all just points of light even in a telescope Okay, right if you look at a planet It's a disc. You can see a circular disc there. Obviously, it's a sphere, spherical. But to a telescope, you see a round thing, okay? William Herschel, I think it was, found a nebula that was round. And he called it a planetary nebula because it looked like a planet like first telescope. Even though he knew it was a gaseous thing and he knew it wasn't a planet, he just was naming things just by what they look like, not for what they are.

And so I'm pissed off that we still have that term for it. Yeah, Herschel, you dumbass. God. Well, the dude discovered other things. So you got to, you know, give him, he discovered infrared light. Right. That's true. I mean, listen, but that was, you know, that was him looking at a table. All right. Where he belonged looking. Okay. We also discovered Uranus. So I know not minus. I don't know who you've been hanging around anyway. That's cool. That's cool. All right. Well, that was very cool. All right. Here we go. This is light blinded fool. And light-blinded fool says, hello, Dr. Tyson. Lord, nice. My name is Robin, and I'm from Vermont. Yay! Love Vermont. Thanks for star talk. My brain is eating very well. My question is, what do you think we'll get to Alpha Centauri first? Missions such as Star Shot or what NASA or what the NASA Advanced Concepts Program is putting together, or will nanotechnology and new propulsion systems outpace these programs if they are launched years later? And could these missions continue beyond Alpha Centauri to more distant stars since they won't be slowing down? Well, so this is why we have telescopes. So we can get a close-up look without actually going there.

I mean, just think about that, okay? You have some object in the distance, you can travel to it or you can whip out a telescope and view it as though you were standing close to it. So the... And even take measurements. And even take measurements. So the urge to be in a place that you could otherwise get measurements via a telescope is reduced simply because we have access to telescopes in multiple wavelengths, with multiple detectors and the like. Okay, so that's my first comment. Second, Starshot, Project Starshot is ambitious. Okay? They're going to have sent posted stamp-sized spacecraft, because so much of this can be miniaturized today. That can

monitor temperature, magnetic fields, thermal fluxes, all in a tiny little posted stamp-sized spacecraft.

that is attached to a huge solar sail, light sail, I should call them. Now, the idea is you launch boatloads of these, and then you have gigawatt lasers on Earth that beam into these light sails to accelerate this very low mass spacecraft. Awesome. And the calculations show, and by the way, if you beam light into a reflective surface, The momentum of the light will be imparted on that surface, and you can accelerate it. The calculations show that beaming at these spacecraft from Earth, even as they increase their distance, you can get the speeds up to 20% the speed of light. I know. I know. Now, now let's do the math. The Alpha Centuary system is four light years away.

So, how long will it take these objects to get there? At 20% of light? 20 years, exactly. Thanks for that good math on the spot. So it'll take 20 years to get there. And so to say, is there some other propulsion, some other technology that we can do within the 20 years that those are in motion that'll just somehow pass them by? I don't think so. I don't think so. How are you getting faster than 20% to speed a light? Yeah, and you're not, I don't see that happening. And just for context, the fastest spacecraft we've ever launched anywhere was the mission to Pluto called New Horizons. And the reason why that was fast is we have a rule, unwritten, but it's definitely a rule. If you're ahead of a space mission, you want to make sure that It's completed before you die. That's an unwritten rule. And so Pluto is very far away. With normal rockets, it might have taken dozens of years. But you make the payload as light as possible, give it low mass, and boost it on the most powerful rockets in the space launch arsenal. This gives you very high acceleration from Earth out.

to Pluto. If instead you redirected that rocket to Alpha Centauri, lest I calculated this, it would take you 40 to 50,000 years to get to Alpha Centauri. That's wild. That's very different from 20 years. Yeah. Good time. Yeah. And that's the fastest thing we've ever launched. So I don't, you know, we're good. This Project Starshot, there's good Google pages on it. You can check it out. And I think it was one, it might have been one of the X prizes. I have to check my memory on that. But anyhow, go check out Project

Starshot. Very ambitious and I think realizable, by the way. We need a few more billionaire investors, but yeah. Very cool. This is Lira, who says, Dr. Tyson, Lord Nice. Vega from Colorado here. Wait, wait. Hold on. How do you spell Lira? L-Y-R-A. So that's the name of a constellation. Okay. Now let's keep going.

Vega says Vega is the name of the brightest star in the constellation Lyra. There you go. So just saying, are we dealing with an alien here? Maybe maybe maybe maybe this is a code. Let's see. Let's see what Lyra says or what Vega says. You talked in previous episodes about there being no functional now. regarding a moving person and a still person looking up in the sky at Andromeda at the same place and time and seeing it years in the past while moving and the present and present while standing. My question is, is there any way we can make a simulation to replicate what true now would be if it were able to exist? Or is our knowledge or understanding unable to comprehend how we could even begin to do that? Yeah, no there's a now for now is a very personal individual thing and By the way, by the way, let me be precise the now as in your clock time. Okay, so You we can all have a now but in what was once synchronized clocks will show up in this case three different nows for you, me and Vega or Lyra, whatever person's name is. Whatever this alien's name is. Right, right, so no, you cannot construct an absolute now. That's kind of the whole, that's what we had to give up when we transitioned from classical physics to relativity physics to modern physics.

What's related to that is we had to give up simultaneity. And that's the same question in a way as asking about now. Because if things are simultaneous, they're happening in the same now. But if there is no absolute simultaneity, that is further evidence that there is no common now in the universe. You can share it now with someone else if you were together with the same speed and the same gravitational field. But otherwise, no, just give up on it. And it's hard to give up classical ways because our senses are forged in, like I said, in the Serengeti to not get eaten by a lion. That's how and why our senses do what they do. Beyond that, you know, the universe is under no obligation to make sense to you. Thank you. There it is.

There you go. All right. Still a good question. I mean, nice to think about this. This is Christian Jeremison. And he says, hello from Elulissat, Greenland. Never heard of it. So one of the ways to visualize gravity is through rubber fabric, a.k.a. bowling ball on a trampoline. And I believe that's known as the rubber sheet, right? So sure. OK. So don't have your bone ball be anything would put a depression in that sheet. Yeah. Right. Does mass stretch our third spatial dimension into fourth or higher dimensions the same way as rubber fabric? If it does, is there still downward gravity in higher dimensions? Oh, yeah. Have some fun with that, Neil, because. Whoa. Whoa. I'm trying to understand it because the.

It's not simply that you have a rubber sheet. Exactly. Because in a rubber sheet, the way we normally think about that, there's like some big source of gravity in the middle of that. Right. And you put a test roller thingy and it rolls around and then it goes into orbit right here. It's to illustrate that the smaller body is falling towards the larger body. That's what it's supposed to illustrate. Correct. Except what that misses, is the fact that everything has distortions in its rubber sheet. So the ball that's rolling around the curved rubber sheet has its own distorted rubber sheet related to it. And so the universe is this juxtaposition of deformed rubber sheets for everything that's in it. From particles out to full galaxies. So we can't think of the mass itself as having some special kind of action on the fabric of space time. It's not. It just has the same action that anything else does. And it corresponds with its total mass. Excellent. Very cool though. But it's good that you asked that because I'm sure when people see that demonstration online, because if you can go on YouTube, many places show that, I'm sure that's the...

intuitive deduction that everybody makes is what he just asked. And in fact, in the original Hayden Planetarium, in the exhibits, we had one of these deformed surfaces. But it wasn't rubber, it was just a rigid deformed surface. And you would take a coin and launch it. And it would roll on its edge around the surface. So you send the coin into orbit around this sort of black hole thingy in the middle. And then we keep your money. That's funny. It was the black hole fundraising mechanism. The only point I'm making there is in that exhibit that deformed sheet was rigid. It was a rigid sheet, a rigid surface. And you weren't thinking that even the coin would have deformations in it as well. So it's easy to see why you would think that

the mass would be something separate and apart from whatever made the deformation in the first place. Right on. OK, let's move on to Rachel Ambrose. That says, hello, Rachel here from Austin, Texas. Neil, are you familiar with Hogue's object?

How statistically improbable is that? From our line of sight here on Earth, we'd see a ring galaxy and then billions of light years behind it. Another ring galaxy that appears inside the first one. Do you think it's more than a coincidence? Maybe some kind of gravitational lensing effect? Okay, I didn't know it had a name. Tell me the name of it again. I know this object, but I didn't know it was named. What are we giving it? She called it Hoag's object. Spell that. H-O-A-G. Hoag's object. Yeah. Okay. I didn't know it had a name. So, let's go back in time. Okay. You look up in the night sky and you see stars. And you ask yourself, I wonder if two stars that are kind of near each other in the sky, are related to each other or they just a chance juxtaposition in space to our line of sight. That's a perfectly natural question. One of the earliest applications of statistics in science, which by the way, happened way later than whole other branches of math were invented. I've written about that in the risk and reward chapter of of the book, Starry Messenger, Cosmic Perspectives. I lament the fact that statistics is one of the last branches of math to be invented, which should tell us something about how ill-equipped our brain wiring is to think statistically about the world. And entire industries exist to exploit that ignorance. And they're called casinos, all right?

Where people, oh, my role is due. Oh, yeah, I think I have feel lucky today. It's like what the brain does to think it is influencing events that are purely probabilistic is sad and tragic. Okay. It is. We have a president who actually bombed a nation based on that same principle. What? Yeah, he said he had a gut. He had a gut feeling. He had a feeling. Yeah, I don't like people's guts. I don't like people's guts. I like people's brains better than I like their guts. Nice. I like that. So can we pose this question mathematically and then arrive at an answer? And yes. So what you can do is you can say, let's take a thousand stars and scatter them randomly on the sky. If you do it randomly, what?

fraction of them will be within a certain angular distance from each other. There's a certain number you would expect if they went up randomly. And random doesn't mean that they're exactly evenly spaced. That is not random. Even if they were random, you could still get a clumping. Yes. In fact, clumpings are expected. Right. This was the problem when people said, oh, he's got a hot hand in a basketball court. Give him the ball. When whatever is your shooting average in a game, you would expect there to be multiple shots in a row that are made. That's normal. It's not that he has a hot hand. That's normal. It's statistically normal, but again, our brains don't allow us to think statistically. We think something magic is going on. We think special kind of luck is happening. So you do this and you know what to expect. Then we looked at how many stars are actually on the sky to a certain telescopic depth.

And then we looked at how many were close by, and there was a statistically significant increase in stars that were close to each other on the sky relative to random. And that was one of the first applications of statistics to assert that we actually have double star systems in the universe. Stars can be born in pairs. And then we said, let's keep going. Stars are born in triplets and quadruplets. open the floodgates to think not only statistically about where stars are in the night sky, but also gravitationally and how you form such objects. Okay, you can do the same thing for galaxies, the same thing. And there are billions of galaxies on the sky. So when you have that many galaxies, things such as alignments of two galaxies, you come to expect them.

You don't go straight to them and say, oh, this is interesting and weird. Oh, what an anomaly. What an anomaly. No, no, no, no, no. Look at the statistical likelihood of it. And we good. We good. We good here. And by the way, we have to be careful because when you do observe subsogics, you don't know initially if they're separate or related. And because they're two coherent things, two rings, you You say, well, if they were on top of each other, could they still maintain their coherence? Their gravity might disrupt their shape. You can ask other sets of questions. Point is, it's not always obvious whether they're connected or not, and you need to do other measurements. Cool. Wow. All right. All right. Still a good question. Yeah. Good question. Thank you for the curiosity there, Rachel. We're all the

better for it. All right. Let's go to Rourke. You know what? I'm going to call him Rory.

His name is Rory now. Your name is Rory now. Rory Kim. Your name is Rory because Chuck cannot pronounce your name. No, it's either Roark or Roark. And that's R-O-A-R-K. Okay, it's work. Work. Okay. I would say work. Not work. Work. Okay. He says, hello, Dr. Tyson. I wonder if the answers to all of our universal questions lay within our ability to see further into geometry and fractal geometry patterns beyond botanical spirals and other phenomena. What if seeing clearly into space isn't what we actually need? Maybe something that helps us see kaleidoscopically, if you will, to uncover those patterns we cannot otherwise see. What is your take on this? You are a gentleman and a scholar. Also, Chuck, if you're reading this, I love you. Oh, that was nice. Okay, so what do you think? Shout out to Chuck. So here's my take on that. I don't want to speak for the whole field, but here's my take. Was it back in the 70s where, but going definitely into the 80s, people started thinking a lot about fractals.

Oh, yeah. Fractals are patterns that you can generate or, and occasionally, find them in nature, where when you zoom, if there's some pattern that you see in a larger view, and then you zoom in, that exact pattern gets repeated. That same pattern. Yeah. Exactly. And you zoom in some more, it repeats again. Oh, they're beautiful. They're beautiful. And you can make beautiful fractal patterns. Yeah. Especially on a computer, because computers are good at that. Here's an example. It's a mild example, but Many plants, I don't know if there's two for all plants. You have to check with a botanist on this. If you look at a tree, and if you look at the angle that a branch comes off from the tree and measure that angle, it's usually upwards a bit, maybe 30 to 60 degree angle from the vertical. Okay. Now that branch will have branches. Correct. Of course they will. This is what I mean by trees branching out. So let's look at the angle that that makes. It's the same angle.

It's the same angle. And look at, if they make a branch, what's that? That's the same angle. So if you just zoom in to this tree, you're getting a smaller tree, even smaller, smaller, you're getting versions of the larger tree as you zoom in. That's just kind of fun to notice if you've never noticed that about plants. I think it's a broader truth for plants that just in general, there may be

important exceptions to it. But if you just look at several plants, you'll see this repeating. And it's fun to notice, by the way. Oh, I got to throw this out there because I only learned it very late in life. If you cut an apple horizontally and you cut into the core and you will see seeds that are radially around the axis of the apple. OK. OK. Do you know how many seed sections there are? No, I don't. No. OK. Neither do I. I don't remember. But it's the same number as the petals on the flower that preceded that fruit. What? I don't know. I thought that was pretty cool. That is cool, man. That's pretty cool. Yeah. All right. That was cool. OK. But I'm not a botanist, so I'm not here to analyze that. I'm just here to just share with you that observation. OK. Anyhow, so fractals, there was the suspicion that Fractals might be something in nature that we would be able to uncover. What makes fractals work is that the same rules that applied on one scale are applying on another scale. Right. It gives you the same thing. Right. In the universe, that's not how it works, okay? There are large-scale forces that do not readily translate to small-scale forces.

Because as you go to smaller in scales, quantum physics kicks in, and that's not the same as Newtonian physics. And none of the rules apply. All the rules are different. So you're not getting the pattern you're looking for. We want to think of patterns because it's easier for us to understand, philosophically easier for us to accept. And our brains are pattern machines. Yes, pattern recognizing Machines like there's never been any. Right. So, so we're good at finding patterns, even when there's no pattern to be discerned. That's how good we are at finding patterns. So true. So true. Which is why I have a pancake in my refrigerator right now, because I don't want to eat Jesus. Oh, he's on the pancake. No, I don't think Jesus shows up on pancakes, shows up on tortillas. Oh, really? I don't think I'm pancakes. No, no, no, no, okay.

So my point is fractals ended up being more entertaining than useful in the sciences. That's my only point. They're entertaining and they're beautiful and you can make fun art. But if you try to use it as a tool to probe nature, it just simply not as useful. And in fact, they tried to model forests using fractal patterns for trees. Okay. Okay. model it in for weather and for, and it just didn't work because the trees are different from each other. Even if one tree, even if one species of tree could be fractal into place, that same fractal

model does not apply to all species of tree. They have to be a different fractal model and how many kinds of, there's thousands or however many species of tree. So it just became messy and awkward.

And so I have not seen much reference to fractals helping scientists decode nature in the last 20 years. Yeah, that's my take on it. If there's a take I've missed, I'd be curious to know how fractals are coming along. But otherwise, I don't see them mattering as much as they we thought they might have. All right. Well, listen, Rick, just go ahead and continue to admire Romanesco broccoli and know that it has. It has no bearing on the universe at all. Roman Eskup Broccoli is fractal food. Yes. Look up a picture of Roman Eskup. Find a picture of it online or buy it. And I'll have you know that in Star Wars Force Awakens, in the bar scene, one of the servers is walking by with a tray of kudates. And one of the objects on that tray is Roman Eskup Broccoli. Oh, cool. All right. Which evidence that that exists in a galaxy far far away. There you go. It's either that or there's one hell of an importer on that planet. So I'm Ali Khan Hemraj and I support Star Talk on Patreon. This is Star Talk with Neil deGrasse Tyson. All right, this is Travis Knapp. This is hello, Dr. Tyson and esteemed funny guest, Travis from Richmond, Virginia with a question inspired by my two year old. Love it. She experiences a year as half her lifetime. While for me, it's only about one 40th. Could there be a cosmic analog? When we see red shifted light from a distant galaxy, we assume the universe was smaller when that light was emitted. Could the same option be right?

But could the same observation be explained by a universe in which time and thus the speed of light passed differently in the past? And is there an observable test that distinguishes expanding space from changing time? Interesting. Love it. Okay. So they are related to each other. Let me get back to time relative to your two-year-old. Of course, your brain is, your life experience is filled by however long you have been, having life experience, right? So you would expect one year to, let's not take a two-year-old because they don't know anything about anything yet, a five-year-old. So one year to a five-year-old is surely bigger in their life experience than one year to a 40-year-old, obviously.

I lived a hard life, man. A lot of people think a five-year-old wouldn't know the ways of the world. But first of all, I quit smoking two years ago. And I just paid down my mortgage. Now, here's something. I quit my job this week. Really mature five-year-old. Yeah, exactly. Let me back into the answer to that question. When I was 14, I was driven with some other people approximately my age from New York City to the Mojave Desert. Wow. To attend an astronomy camp where I lived nocturnally for a month. Wow. The trip, we drove nonstop and the trip took two days and five hours. Ugh. I didn't correct for time zone change. So it's two days and one hour. Okay. Okay. Is that right? Or is it the other way? How have you correct that for time zone? Point is, it took two days. When we arrived, it seemed like we had left New York a week earlier because those two days were so filled.

with, oh wow, that's the St. Louis Arch. Oh my gosh, we're crossing the Mississippi River. Oh, is this Tennessee? Oh, is this Texas? My life was just, my visual mental awareness of the world was so full that my brain wouldn't allow me to say, oh, that all happened in the last 48 hours. My brain forced me to think that that was spread over many more days than it actually was. because the density of life experience was so high, was higher than anything I'd experienced in my life. So that was my mind's attempt to make sense of how much, how action filled those two days were. Anyhow, that's purely psychological. And you can know this because you can bring measuring devices, clocks and things. And no, it didn't take me a week to get to California. It did only take two days. All right.

In the expanding universe, it does affect time. It does. And I am proud to say I am a co-author on the first paper, research paper, to establish that fact. Okay? I forgot it might have been as many as eight or nine authors on this paper. And the lead author, the lead author, would later win the Nobel Prize for the discovery of dark energy, for code discovering dark energy in the universe. So the lead author on that paper, his name is Brian Schmidt, who studied the, as we say, high redshift supernovae. These are stars that explode in galaxies out to the edge of the universe. And if you do that, you can probe metrics of the universe, because all the supernovae have a certain common properties.

So it's like a yardstick, a free yardstick throughout the universe. And so he would ultimately share the Nobel Prize with Adam Rees for this discovery. And both those groups, the high redshift supernova groups, one was on the East Coast, one was on the West Coast. Brian Schmidt is now, last I checked, he was like provost in Australia, Australia National University, I think it is. But anyhow, so I'm on a research paper that showed that if you know when a star blows up a supernova, it gets brighter and then it gets dimmer at a very predictable rate. We know this, okay? That's why they make good yardsticks, standard candles, as we've called them. There was a high redshift supernova that didn't fit this light curve. We call it a light curve. And we said, hmm, is this a different kind of supernova? Or what? Because This supernova hails from when the universe was half its current size, and the universe was expanding as it still is. So what's up with that? Turns out, if you put in the time dilation expected for the expansion of the universe at the time of that galaxy's light, it stretches the light curve to be exactly on the curve we measure. Holy shit.

It was it was a direct evidence of the not only a shift in the expanding universe But what effect that has on time? Because the light curve was stretched out right the We observing that light curve it took longer for that light curve to execute its rise and fall than a nearby supernova out exactly the rate that you'd expect from the expanding universe and so So I'm proud to be on there. I supplied some data, some supernova data that he compiled to make the full paper. So it's his effort to come up with that result. I was a cog in a much larger wheel to make that happen. So the point is, yes, time is affected. And we see that in phenomena, in galaxies and other phenomena, in those places and at those times in the early universe.

the speed of light as you measure it is the same no matter what. No matter what. That's not what changes here. Right. But the time keeping is what changes. And so there you have it. That's wild. All right. This is Kyle Holman. Kyle says, hello, Dr. Tyson. Lord, nice, Kyle from Fort Myers, Florida here. Hey, Fort Myers. First time Patreon member. Welcome aboard. Welcome to the Start Talkiverse. Nice. There we go. What might the night sky have looked like when the cosmic microwave background radiation was cosmic visible light background? Oh, okay. So the cosmic background today is microwave. It's microwaves, right? Microwaves, okay. They

microwave Photon is like one centimeter long or less. Two millimeters up to a couple of centimeters. It's the length, the physical length of the wavelength of the photon. Okay. That has stretched in the expanding universe since the cosmic microwave was first formed. Okay. So we ask, what was the temperature of the universe when it first formed?

Because today the temperature that corresponds with microwaves is three degrees Kelvin absolute Kelvin, okay The temperature of the cosmic microwave background when it formed was about 3,000 degrees It turns out that there's an exact one-to-one relationship between how much the wave has stretched and how much bigger the universe is so in the early universe when its temperature was 3,000 degrees relative to three degrees today. It means the universe was one one thousandth its current size because over those years it's expanded a thousand fold going from the wavelength of light at three thousand degrees to the wavelength of light at three degrees. And what is the wavelength of light at three thousand degrees? It's visible light. So at one point the whole universe was just a beautiful like heavenly light.

Yes, a heavenly glow. But it would be like the surface of the sun, but everywhere. And if you had good measuring devices, you can detect that some parts were slightly cooler and slightly warmer than others. And that you'd be immersed in that field. And you wouldn't need special telescopes to see it. So when you say night sky, people often think of stars, suns, moons and planets. Of course, none of that had formed yet because the universe is still trying to get its act together. Mm-hmm. Mm-hmm. Those early moments. Yeah. Oh, that's cool. Yeah, that is really cool. Damn. Yeah, all right. So you would have seen that, you'd have seen that and just been like, somebody turn that down. Yes, exactly. There'd be no, there'd be no, look, you couldn't, you couldn't get away from it. Right. It'd be everywhere. It's omnipresent, just light. And let me tell you how bad it actually would be.

If you said, oh, there's some shade over there, and you go under the shade, all of that energy will heat the thing that's shading you, and then it will glow the same temperature as the background. Wow. OK, so you wouldn't... It'd be inescapable. Inescapable. Wow, very cool. Oh, man, that was good, man.

Here's a quick thing. I don't know if you've done this. Have you ever done pottery where you have a kiln? Have you ever done this? I wouldn't quite call it pottery. I mean, I took some clay and some water and supposedly shaped it into something. And then we put it in a little oven. And then when it came out, I was like, throw that shit away. Okay. So a kiln, you know, where you fire up the kiln and you, it sets your clay.

Okay. That is glowing hot. Yes, it is. Okay. It is glowing hot. It's hotter than red hot, a little hotter than red hot. All right. So there it is. That's about 3000, call it 3000. It might be as low as 2000 degrees, but let's call it 3000 degrees for the moment in the kiln. There it is. If you look inside, it's just glowing. Right. You can't see the edges of the walls of the kiln because it's all glowing at you. That's right. That's true. Now, you take your pottery, put it in there. What is the temperature when you put it in there? What does it do or what is it? It says... What is the temperature of your pottery when you put it in? Oh, it's whatever the atmospheric temperature is. Room temperature, okay. So now you put it in, close the thing, and that pottery gets hotter and hotter and hotter. You can watch this happen. It gets hotter and hotter and hotter.

until it reaches the same temperature as the glowing walls. And then it disappears because it is the same glowing temperature surface as everything else. Maybe if you move your head a little bit, you can catch some edges of it, but compared to what it was when you first put it in, when it's a discrete object, glowing at a room temperature rather than the temperature of the kiln itself, so it's how to make something disappear. in the background. And one last example, completely obscure, but I gotta put it out there, in the film, which I highly recommend, because it's very, it's masterfully crafted and very sexy and it's a caper. It was the remake of the Thomas Crown affair. Okay? Okay.

It's a heist in the Metropolitan Museum of Art, except they didn't agree to get their name mentioned in the credits. I think they don't want people heisting their paintings. Anyhow, anyhow, the thieves waited for the hottest day of the year. Okay. Then cut the air conditioning to the rooms so that the air got up to the temperature of the walls. And so the camera, they have infrared cameras. When that happened, the infrared camera could not detect

the paintings from the walls, from the air in front of it. And the whole, the whole scene went, I don't know, blank. The whole scene went uniform. Right. And yeah, because all glowing at the same temperature. that's pretty cool um if you put the pottery in the kiln and look in the window when it disappears if you listen very very closely you can hear the pottery saying I'm referring only to the light waves, not the sound waves. That's a different conversation. Oh, another thing, by the way, if the temperature in the rooms go up to like 100 degrees, because it's like the hottest day of the year and the walls go to 100 degrees, what's your body temperature? 98 points, whatever. Yeah, that's so that's near 100 degrees. So all of this just blends. It turns out we have a higher.

density of infrared light coming off of us than the air does, but it still makes it much harder to figure out what's going on when everything is glowing to you at the same temperature. That's all. So that's a lot of obscure everyday references to something that started as a cosmological, a brilliant cosmological question about the cosmic microwave background. Nice. Well, thanks, Kyle. You got it. Yeah. I think that's all the time we have. Oh, well, this was fun. Another installment of Cosmic Queries. Grab bag. Gumball. Gumball. Gumball. Gumball. Gumball. Gumball. Gumball. Gumball. Gumball. Gumball. Gumball. Gumball. Gumball. That's correct. On our YouTube channel. Oh, thank you. Yeah. On StarTalk's YouTube channel. That's right. And you recorded that and posted it New Year's Eve or New Year's Day this year. New Year's Day. New Year's Day. So that's right. It's you just telling it like you see it. Yeah, Chuck and I are just smart enough on the StarTalk YouTube channel. Go and download it and share it with your friends. It's free. Got it. That's all the time we have, Chuck. Wow, man. This was good. I am Neil deGrasse Tyson. You're a personal astrophysicist. Keep looking up.